

ACTIVITY 20 — LEARNER WORKSHEET**Earned Value Management and Schedule Compression**

TYPE	DURATION	MAPS TO	TOPIC
Workshop	60 minutes	LO5	5 - Monitor and Control the Project

What you are doing

Contoso Learning Pte Ltd is a training provider with 14,000 active learners. Its course registration portal is 9 years old. Registration takes a median of 11 minutes and 34% of attempts are abandoned. Learner communications consume 24 staff-hours per week across 3 administrators. 62% of learner traffic is mobile, against a portal that is not responsive. Legacy consent screens do not meet current PDPA requirements. The project modernises course registration and learner communications.

What you will produce: evm calculations, forecast, compression decision — submitted to artifacts/.

What you need before you start

- Lab 13 cost baseline and time-phased PV
- Lab 12 critical path and crash costs
- Lab 03 escalation thresholds
- Lab 18 status report formats

Your steps

1. Take the Contoso baseline and actuals. This is the real time-phased cost baseline from Lab 13, together with the measured EV and AC at each fortnightly reporting point. The project is at the end of week 14, which is the end of sprint 7 of 12.
2. Compute the variances and indices at week 14. Work these by hand before checking. Use the cumulative figures - EVM is always cumulative unless a question explicitly says "this period".
3. Forecast the outcome. Compute all three EAC variants, then choose the defensible one.
4. Analyse the trend, not the snapshot. A single period tells you little. Compute CPI and SPI for every period and look at the direction.
5. Check the escalation threshold. Go back to your Lab 03 escalation thresholds and apply them. This is the step that converts analysis into required action.
6. Decide: crash or fast-track. The Board asks how the schedule position (SPI 0.929) will be recovered. You have two levers, and they fail in different ways.
7. Write the variance report. Use the Lab 18 sponsor format. State the position, the cause, the action and the ask.
8. Answer the exam-style scenarios

Your working space

Complete every field. Keep this worksheet — it is your reference for the rest of the course.

Step 1 — Take the Contoso baseline and actuals

Step 2 — Compute the variances and indices at week 14

Step 3 — Forecast the outcome

Step 4 — Analyse the trend, not the snapshot

Step 5 — Check the escalation threshold

Step 6 — Decide: crash or fast-track

Step 7 — Write the variance report

Step 8 — Answer the exam-style scenarios

Deliverable checklist

- 20-evm-calculations.md - CV, SV, CPI and SPI at week 14 with the working shown, plus the full seven-period CPI/SPI trend table.
- The baseline validation check proving cumulative PV at period 12 equals BAC.
- 20-forecast.md - all three EAC variants computed, a reasoned choice of which applies with justification from the trend, plus ETC, VAC and both TCPI figures.
- Your Statistics tool output for the CPI series: mean, standard deviation, trend slope, and the extrapolated period-12 CPI.

- The escalation decision with the threshold arithmetic, stating whether you escalate on actual or forecast variance and why.
- 20-compression-decision.md - the crash analysis identifying the optimal 5-day recovery, its cost, the reserve consequence, and the documented reason for rejecting the fast-track option.
- The full variance report in sponsor format with three costed options and a recommendation.
- Written answers to the three scenarios.